

IIT MADRAS BS IN DATA SCIENCE & ELECTRONIC SYSTEMS

Deep Learning & Generative AI

Weeks 1 & 2 • Complete Practical Compendium

Compendium Table of Contents & Structure

01

PyTorch Fundamentals

Tensors, Tensor Attributes, Mathematical Operations, Reshaping, Slicing, Matrix Multiplication, Device Agnostic Execution & CUDA Acceleration.

02

PyTorch Workflow & End-to-End Modeling

Data Ingestion, Custom nn.Module Architecture, Forward Propagation, Loss Computation, Optimizers, Epoch Training Loop, Evaluation & Checkpointing.

03

Customer Churn Prediction (ANN Binary Classification)

Tabular Data Wrangling, Feature Scaling, Multi-Layer Perceptron (MLP), Binary Cross-Entropy Loss, Precision-Recall Curves & Production Thresholding.

04

Fashion MNIST Classification (Multi-Class Vision)

Torchvision Image Datasets, Mini-Batch DataLoader Pipelines, Non-Linear Vision MLPs, Cross-Entropy Loss Optimization & Multi-Class Confusion Matrix Analysis.